Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/21/2017
PI Name:	Stuster, Jack W. Ph.D.		
Project Title:	Behavioral Issues Associated with Long Duration Space Missions: Review and Analysis of Astronaut Journals		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	New address per PI (12/2012); previous address--301 East Carrillo Street, Santa Barbara, CA		
Project Type:	Flight	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/02/2011	**End Date:**	05/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Element change to Human Factors & Behavioral Performance; previously Behavioral Health & Performance (Ed., 1/18/17) NOTE: New end date is 5/30/2017 per PI (Ed., 12/2/14) NOTE: Extended to 9/30/2015 per PI and NSSC information (Ed., 1/27/14) NOTE: Start date changed to 3/2/2011 (from 10/15/2010) per C. Guidry/JSC (Ed., 2/7/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX11AG22A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This study is a continuation of the Journals Flight Experiment, which concluded successfully in April of 2010 with a report that documented the systematic analysis of confidential journals maintained by ten astronauts who were members of two and three-person crews onboard the International Space Station (ISS); the second phase of the study involved the analysis of journals maintained by ten additional astronauts while serving as members of six-person crews. The objective of the study is to provide data-driven recommendations to inform decisions concerning the priorities that should be placed on the various behavioral issues to prepare for and manage future space expeditions. Study results can be applied to improve selection, training, support, scheduling, and the design of equipment and procedures. The specific objectives are to 1) collect data concerning the relevant behavioral factors from astronauts while serving as one-year crewmembers aboard the International Space Station (ISS); 2) Compare the experiences reported by members of one-year crewmembers to those of shorter duration who participated in earlier journals studies; 3) Assess the effects of changes implemented since the previous journals study (e.g., procedures); 4) further test the third quarter phenomenon; and 5) provide a safe and constructive outlet for the frustration and irritation that usually accompany isolated and confined living. The ultimate objective of the study is to provide data-driven recommendations to inform decisions concerning the priorities that should be placed on the various behavioral issues to prepare for and manage future space expeditions. The Journals Study will also perform a systematic analysis of journals content from one-year ISS astronauts. Study results can be applied to improve selection, training, support, scheduling, and the design of equipment and procedures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Conditions on Earth that are characterized by isolation, confinement, and austere or hazardous duty could benefit from the results of the Journals flight experiment.
Task Progress:	Data were analyzed and the final report documenting both Phase 1 and Phase 2 of the Journals Flight Experiment was prepared. Study results were presented at the annual Behavioral Health and Performance (BHP) meeting, to the Astronaut Office, and to groups of flight controllers and flight directors. The Journals Flight Experiment was completed under Cooperative Agreement NNX11AG22A for the Behavioral Health and Performance Element, Human Performance Program, NASA's Johnson Space Center. The study described in the final report (NASA/TM-2016-218603; see Bibliography section) was a continuation of work performed between 2003 and 2010 that analyzed the personal journals of 10 astronauts who were members of 2 and 3-person crews onboard the International Space Station (ISS); the research performed under NNX11AG22A was conducted between 2011 and 2016 and analyzed the journals of 10 astronauts who were members of 6-person crews, which enabled comparisons to be made. Both phases of the study are based on the premise that the introspective accounts of individuals who are living and working in stressful or unusual environments can provide useful information about the factors that affect individual and group performance under those conditions. Study participants included men and women; scientists, engineers, physicians, and pilots; and civilian and military personnel. The 10 participating astronauts in Phase 1 (members of 2 and 3-person crews) devoted a total of 705 sessions to the Journals Experiment and produced 4,247 parsed statements composed of nearly 285,000 words; the 10 participating astronauts in Phase 2 (members of 6-person crews) devoted a total of 551 sessions and produced nearly 4,000 parsed statements composed of 209,000 words. The astronauts' combined journals are equivalent to two, 1000-page books and document ten person-years of living and working in space. The methods used to analyze journal content were developed during space analog research and are based on the assumption that the more an astronaut writes about an issue, the more important the issue is to the astronaut at the time of writing. Each journal entry was parsed into statements and the statements were assigned to a primary behavioral category (secondary and tertiary categories were indicated for many statements). Journal statements also received a code to indicate whether the statement was positive, negative, or neutral in its tone or content and then a metric called Net Positivity/Negativity (NPN) was derived by subtracting the proportion of negative entries from the proportion of positive entries (e.g., by behavioral category, subject). Statements were coded for mission day to enable temporal analysis. The first level of analysis identified the relative salience of 24 major categories of issues with behavioral implications. The top four behavioral categories during Phase 1, Work, Outside Communications, Adjustment, and Group Interaction, received 59% of all primary, secondary, and tertiary category assignments of journal statements. Adding Recreation/Leisure, Equipment, Event, Organization/Management, Sleep, and Food to form a list of the top 10 categories accounts for 88% of all category assignments. Nine of the top ten Phase 1 categories were in the top ten during Phase 2, with only a slightly different order: Adjustment, Work, Outside Communications, Group Interaction, Equipment, Recreation/Leisure, Food, Event, and Organization/Management; Exercise replaced Sleep during Phase 2. The top four categories accounted for 61% of all statements during Phase 2 and the top ten accounted for 86% of the total. The second-level of analysis identified more than 100 subcategories within the major categories and calculated their relative frequencies and temporal distributions. The third level of analysis focused on the tone of entries as an indicator of specific problems and general morale, using the NPN metric; results of the NPN analysis support hypotheses concerning a third quarter phenomenon, which is expressed as a decline in morale during the third quarters of expeditions, regardless of duration. Responses to questions asked before, during, and after the expeditions suggest that living and working onboard the ISS is not as difficult as the astronauts anticipate before starting their six-month tours of duty. Astronauts reported that they benefited personally from writing in their journals and recommended continuation of the research. Examples of statements from the journal entries and graphic representations of data are used in the final report to illustrate the quantitative analyses. Together, the statements and analyses provide a comprehensive description of ISS operations from the crew perspective. Example journal statements included in the report concern activities, events, and observations and reflect the normal ups and downs of daily life onboard an orbital space station. It is important to note that no single journal statement, nor even a subset of statements, accurately describes an entire expedition, but rather, the most salient issue at the time the entry was made. The results show that NASA's efforts concerning interpersonal issues, teamwork, psychological support, and leadership have been effective. Operational implications of study results are summarized and then presented in the form of specific recommendations to facilitate living and working in space, whether onboard the ISS, a spacecraft bound for an asteroid, or an interplanetary ship.

Bibliography Type:	Description: (Last Updated: 11/13/2019)
NASA Technical Documents	Stuster J. "Behavioral Issues Associated With Long Duration Space Expeditions: Review and Analysis of Astronaut Journals Experiment 01-E104 (Journals) Phase 2 Final Report." Houston, Tex. : NASA Johnson Space Center, 2016. NASA technical memorandum TM-2016-218603. , Apr-2016